

## **EKSAMENSOPPGÅVE/EKSAMENSOPPGAVE**

Emnekode: MAS 149

Emnenamn/Emnenavn: STATIKK OG FASTHETSLÆRE

Utdanning/kull/klasse: Diverse

Dato: 16.06.2025

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Eksamensform: Skriftlig

Eksamenstid: 2 timer

Tal på eksamensoppgåver/Antall eksamensoppgaver: 4

Tal på sider (medrekna denne)/Antall sider (inkludert denne): 10

Tal på vedlegg/Antall vedlegg: 2

Tillatne hjelpemiddel/Tillatte hjelpemidler: Enkel kalkulator

(ikke programmerbar)

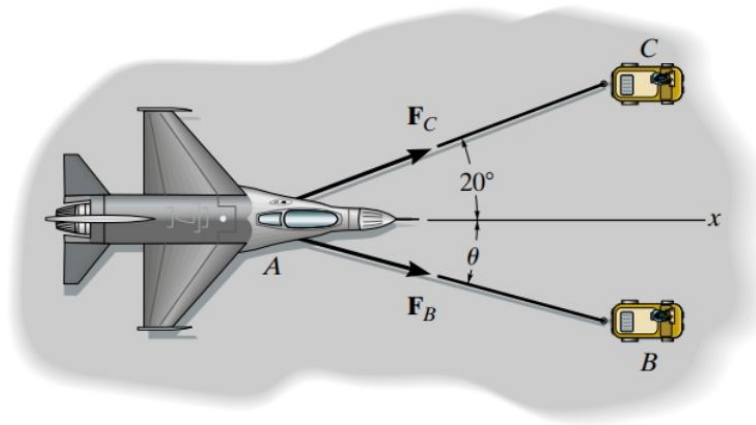
Fagansvarleg/Fagansvarlig: Javier de Elio Medina (Haugesund)



## English version

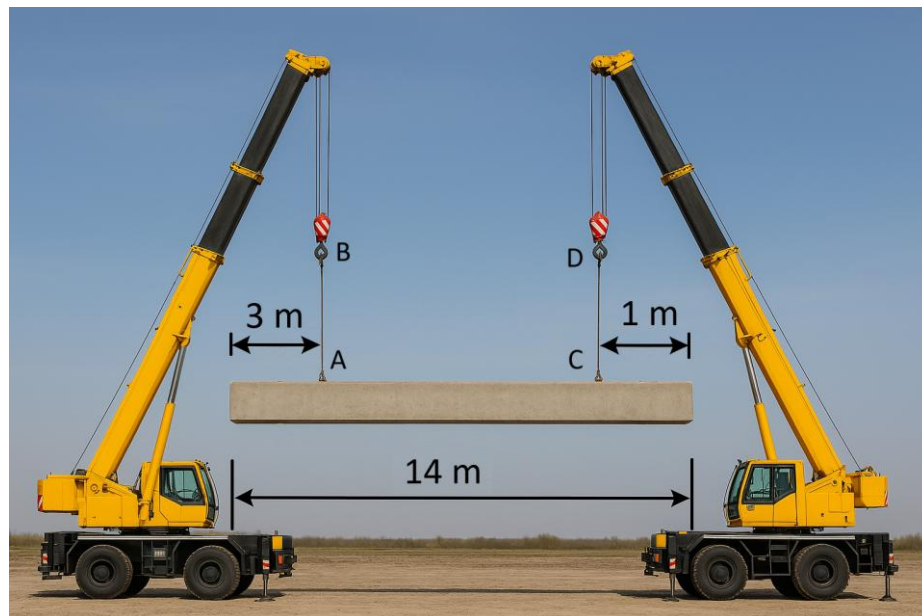
### Exercise 1 (25%):

Determine the magnitude of the two towing forces  $F_B$  and  $F_C$  if the resultant force has a magnitude  $F_R = 15 \text{ kN}$  and is directed along the positive  $x$  axis. Set  $\theta = 10^\circ$ .



### Exercise 2 (25%)

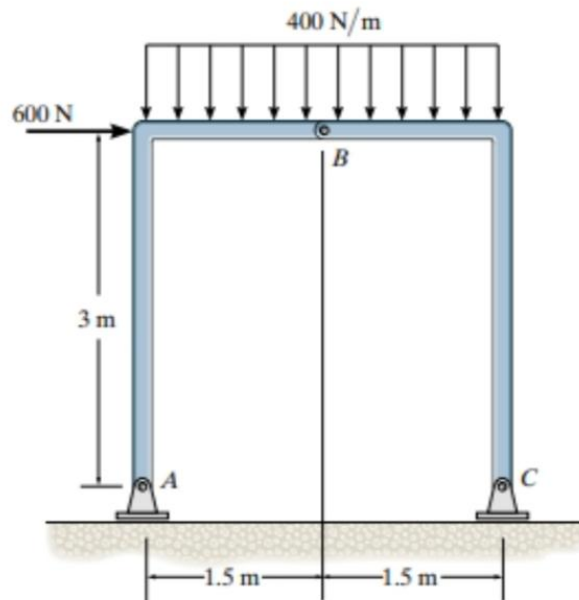
A homogeneous concrete beam of 14 m and 10 t is being lifted by two cranes as shown in the figure. Determine the tension in cables AB and CD.



Note: Figure generated with ChatGPT

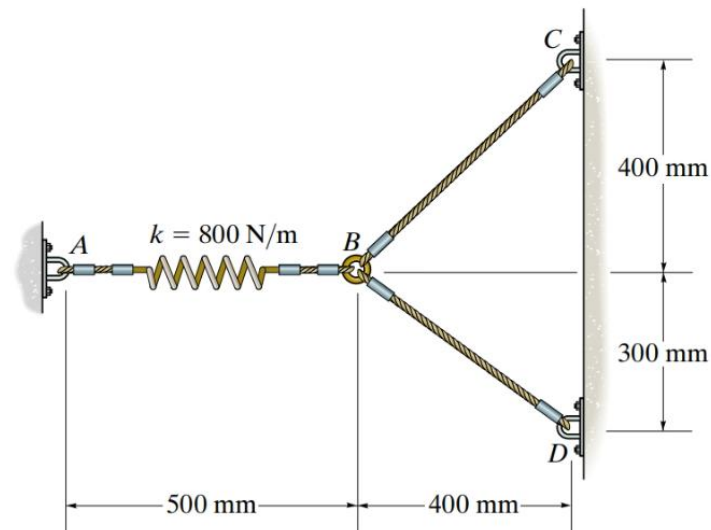
**Exercise 3 (25%)**

Determine the horizontal and vertical components of the forces that pins A and C exert on the frame.



**Exercise 4 (25%)**

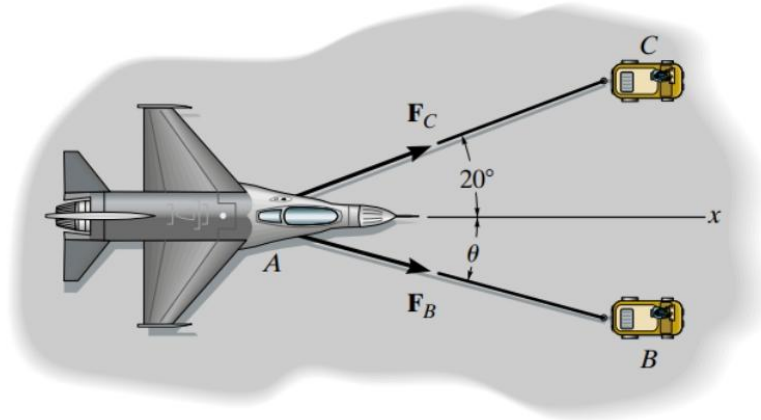
Determine the forces in cables BC and BD required to hold the spring in the position shown in the figure. The spring has a stiffness of  $k = 800 \text{ N/m}$  and an original length of 200 mm.



## Bokmål versjon

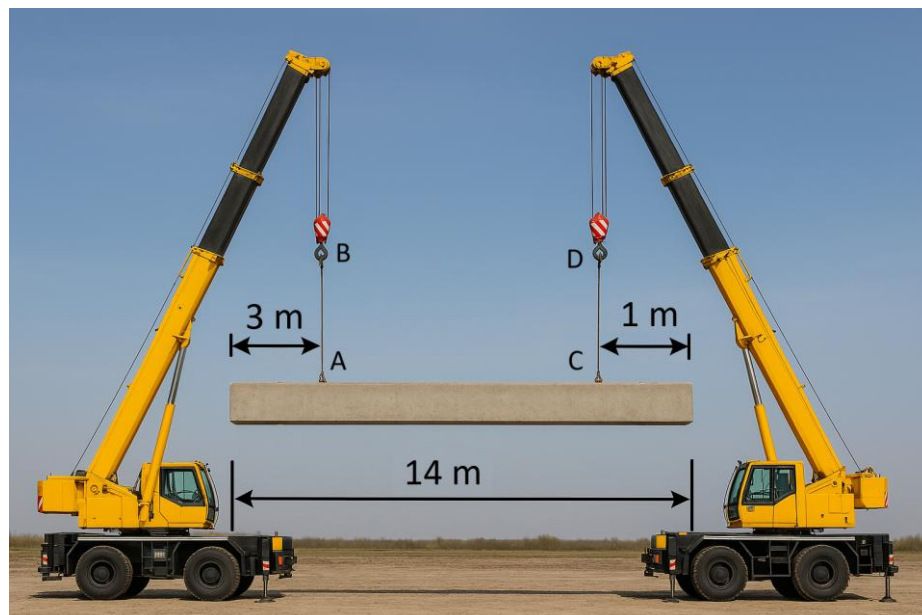
### Oppgave 1 (25%):

Bestem kreftene  $F_B$  og  $F_C$  dersom resultantkraften  $F_R$  har en størrelse på 15 kN og er rettet langs den positive x-aksen. Sett  $\theta = 10^\circ$ .



### Oppgave 2 (25%)

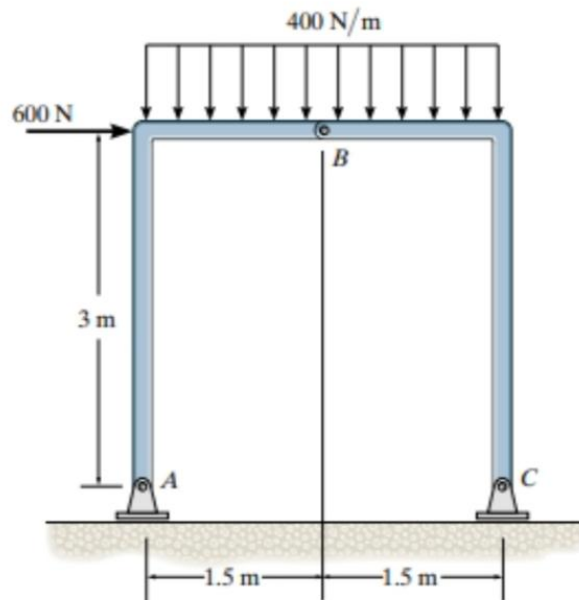
En homogen betongbjelke på 14 m og 10 t løftes av to kraner, som vist i figuren. Bestem spenningen i kablene AB og CD.



Note: Figuren er generert av ChatGPT

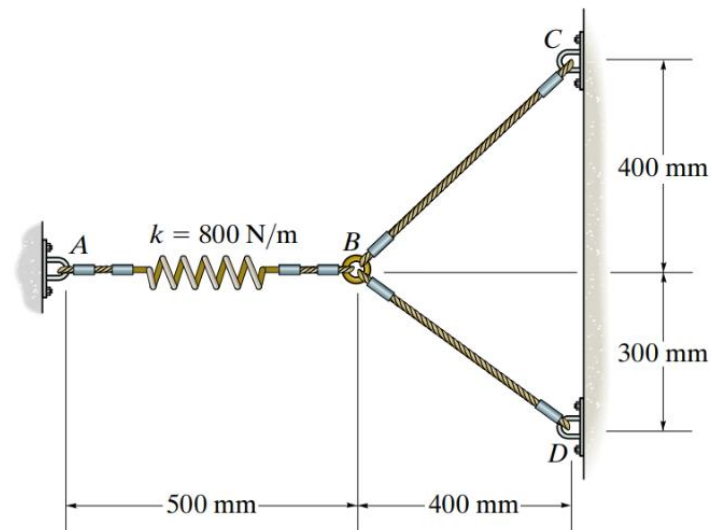
### Oppgave 3 (25%)

Bestem de horisontale og vertikale komponentene av kreftene som pinnene A og C utøver på rammeverket vist i figuren.



### Oppgave 4 (25%)

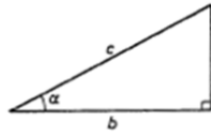
Bestem kreftene i kablene BC og BD som er nødvendige for å holde fjæren i posisjonen vist i figuren. Fjæren har en fjærstivhet,  $k = 800 \text{ N/m}$ , og en opprinnelig lengde på 200 mm.



## TRIGONOMETRI

Pytagoreiske  
læresetning

$$c^2 = a^2 + b^2$$



$c$  hypotenus  
 $a$  |  
 $b$  kateter

Trigonometriske  
funksjoner

$$\sin \alpha = \frac{a}{c}$$

$$\cos \alpha = \frac{b}{c}$$

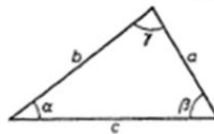
$$\tan \alpha = \frac{a}{b}$$

Sinussetningen

$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}$$

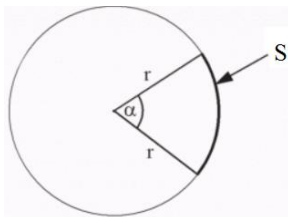
Cosinussetningen

$$a^2 = b^2 + c^2 - 2bc \cos \alpha$$



Tangenssetningen

$$\frac{a-b}{a+b} = \frac{\tan \frac{\alpha-\beta}{2}}{\tan \frac{\alpha+\beta}{2}}$$



$$S = \alpha \cdot r \quad \alpha [\text{rad}]$$

## Spring:

$$F = k \cdot s$$

$$s = l - l_0$$

$k$  = spring constant or stiffness [N/m]

$l_0$  = undeformed length [m]

## Geometric Properties of Area Elements

## Material Property Relations

Poisson's ratio

$$\nu = -\frac{\epsilon_{\text{lat}}}{\epsilon_{\text{long}}}$$

Generalized Hooke's Law

$$\epsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)]$$

$$\epsilon_y = \frac{1}{E} [\sigma_y - \nu(\sigma_x + \sigma_z)]$$

$$\epsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)]$$

$$\gamma_{xy} = \frac{1}{G} \tau_{xy}, \gamma_{yz} = \frac{1}{G} \tau_{yz}, \gamma_{zx} = \frac{1}{G} \tau_{zx}$$

where

$$G = \frac{E}{2(1 + \nu)}$$

Relations Between  $w$ ,  $V$ ,  $M$ 

$$\frac{dV}{dx} = w(x), \quad \frac{dM}{dx} = V$$

## Elastic Curve

$$\frac{1}{\rho} = \frac{M}{EI}$$

$$EI \frac{d^4 v}{dx^4} = w(x)$$

$$EI \frac{d^3 v}{dx^3} = V(x)$$

$$EI \frac{d^2 v}{dx^2} = M(x)$$

## Buckling

Critical axial load

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}$$

Critical stress

$$\sigma_{cr} = \frac{\pi^2 E}{(KL/r)^2}, \quad r = \sqrt{I/A}$$

Secant formula

$$\sigma_{max} = \frac{P}{A} \left[ 1 + \frac{ec}{r^2} \sec \left( \frac{L}{2r} \sqrt{\frac{P}{EA}} \right) \right]$$

## Energy Methods

Conservation of energy

$$U_e = U_i$$

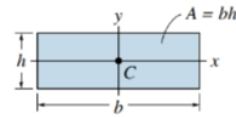
Strain energy

$$U_i = \frac{N^2 L}{2AE} \quad \text{constant axial load}$$

$$U_i = \int_0^L \frac{M^2 dx}{2EI} \quad \text{bending moment}$$

$$U_i = \int_0^L \frac{f_s V^2 dx}{2GA} \quad \text{transverse shear}$$

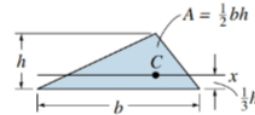
$$U_i = \int_0^L \frac{T^2 dx}{2GJ} \quad \text{torsional moment}$$



Rectangular area

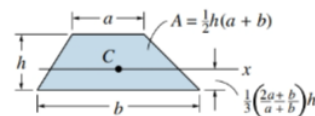
$$I_x = \frac{1}{12} bh^3$$

$$I_y = \frac{1}{12} hb^3$$

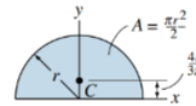


Triangular area

$$I_x = \frac{1}{36} bh^3$$



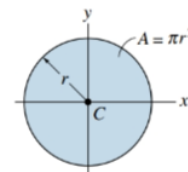
Trapezoidal area



Semicircular area

$$I_x = \frac{1}{8} \pi r^4$$

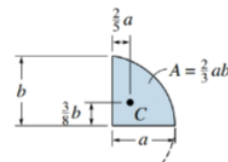
$$I_y = \frac{1}{8} \pi r^4$$



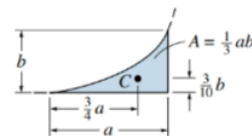
Circular area

$$I_x = \frac{1}{4} \pi r^4$$

$$I_y = \frac{1}{4} \pi r^4$$



Semiparabolic area



Exparabolic area